USEFUL LINKS

Flood mapping tool (FMT): *https://floodmapping.inweh.unu.edu*

Mapping of flood areas using Landsat with Google Earth Engine Cloud platform: *https://www.mdpi.com/2073-4433/12/7/866*

Presentation on the new tool: *https://bit.ly/2XZi8ld*

Hamid Mehmood, GIS and remote sensing specialist, interview: *https://youtu.be/ijvZZiVDLyA*

In the next release of the tool, the team plans to increase the resolution of the open-access version by using data from multiple satellite sensors (harmonised Landsat Sentinel-2 data). A paid version of the tool with sub-metre accuracy is also in development, which caters to the need of the commercial sector (insurance, real estate, agriculture).

Role for GIS and remote sensing

To mitigate the impact of flooding, experts who created this solution are pushing governments to do three things: (1) Improve the accuracy of flood maps to capture the true extent of historical floods, rivers, streams, and water bodies; (2) Improve risk mapping and introduce policies to reduce risk; and (3) Invest in the infrastructure to reduce risks at the community level.

"In the last 5-10 years, GIS and remote sensing has turned into a cross-cutting technology. GIS-remote sensing has evolved a lot. It integrates data from UABS (unmanned aerial base stations), sensors, and IoT (Internet of Things), and starts using the cloud computing platform. I think it will make a lot of contribution in meeting these global challenges," said Dr Mehmood, indicating his optimism about the field he works in. END

Facts and figures

- Floods in the past decade have impacted the lives of over half a billion people, which is one out of every sixteen persons on the planet, the UNU says. Those most affected are in the low- and middle-income countries.
- The damage due to these floods has cost nearly US $500 billion, roughly equal to the GDP of Singapore.
- Some 1.5 billion people across the globe — more than Europe's population — live at the risk of exposure to intense flooding.
- A majority of flood forecasting centres in flood-prone countries lack the ability to run complex flood forecasting models.
- Wikipedia's list of history's deadliest floods list 211 events, 103 of which have occurred from 1985.
- Cyclones, floods, and droughts — called water-related disasters, or WRDs — account for 90 per cent of natural disasters. Since 2000, over 5,300 WRDs have killed over 325,000 persons and caused an economic loss exceeding US$ 1.7 trillion globally. Floods make up 54 per cent of all WRDs.
- Since the start of 2020, South Asian floods have impacted over 17.5 million people, caused over 1000 deaths, and cost billions of dollars.
- In times of climate change, "far more intense and more frequent" floods can be expected, says the UNU. Urban areas are likely to grow 80 per cent in Africa and Southeast Asia, and the tool could suggest flood-safe locations for housing and industry, and overall urban planning.

"This is a really interesting tool, to understand where floods have happened and how they have affected populations. What will make it more interesting and useful is the temporal aspects and land use imagery.

"If it can be mapped over Google Maps, which is open source, it will help lay people who can read maps understand where floods have happened and where they are likely to happen in the future.

"Rainfall data could make it even more useful to match floods to the quantity of rain over a particular period. For example, if there has been heavy rain on Day 1 in the catchment area, there is a flood on Day X in the downstream area. That could help understand and plan for evacuations or countermeasures. It will also help to understand where catchment area treatments will have the most impact in mitigating floods.

"Many Indian water experts have been mapping aquifers, watersheds and river basins. But there is nothing like this in India. Given our vulnerability to floods, this tool could be really useful here."

*—**Nitya Jacob**, water policy analyst and author of 'Jalyatra, Exploring India's Traditional Water Management Systems'*

By: Frederick Noronha

The author is an independent journalist and publisher, and has written on FOSS for almost two decades.